Aleksandra Winkler

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Education

University of Utah, BS in Physics with an emphasis on Astronomy — August 2024 - May 2027

GPA: 3.9

- **Deans List 4 Semesters**
- **Coursework:** Cosmology, Observational Astronomy, Foundations Astronomy, Relativity, Kinematics, Mechanics, Electrostatics, Quantum Mechanics, Thermodynamics, Differential Equations, Linear Algebra, Complex Variables, PDE's

Weber State University — August 2023 - April 2024

GPA: 3.9

- **Deans List 2 Semesters**
- **Coursework:** Calculus Series (I, II, III), Introductory Astronomy, College Algebra, Trigonometry
- **Awards:** Utah NASA Space Grant Consortium awardee, WSU STEM Underrepresented Scholarship, Doris Parker Women in Physics Scholarship

Experience

Astrophysics Research, Mentored by Dr. Catherine Manea at the University of Utah — January 2026 - Present

- Exploring multiple wide binary systems and their chemical compositions by analyzing high-resolution stellar spectra
- Contributing to the ongoing research as to why the chemical makeup of two stars in a wide binary can be widely differing when we should expect them to be chemically homogeneous
- Differential analysis methods using Python and BACCHUS (Brussels Automatic Code for Characterizing High-accuracy Spectra) allow us to parameterize star pairs up to a very high precision

Mathematics Research, Mentored by Dr. Kurt Vinhage at the University of Utah — April 2025 - June 2025

- Collaborated with 3 other researchers and conducted research on what types of matrices are included in the Lorentzian Group and utilized SageMath and LateX to communicate our results to our peers
- Applied extensive linear algebra and differential geometry concepts to classify Lorentz transformations and showed how this could be interpreted into a physical representation such as a light cone
- Conceptualized how the tilt of the light cone can be used to describe the effect black holes have on space-time through coding simulations

Physics Lab Learning Assistant, University of Utah — August 2025 – Present

- Helped peers discover core physics topics during their laboratory time and answer questions to my fullest ability
- Clarified confusion on MatLab code for peers who are brand new to the language and concepts
- Provided an encouraging space for fellow undergrad students to ask questions and feel very open to learning

Professional Ballet Dancer, Ballet Metropolitan — June 2022 - May 2023

- Collaborated with dancers, directors, and repetiteurs to produce ballet performances multiple times a year
- Self corrected mistakes in technique and style to improve overall as a dancer, learned to be very detail oriented
- Familiarity with being involved in a high stakes environment and determination towards achieving a major goal

Skills

Programming & Tools: Python (NumPy, SciPy, Matplotlib, Astropy), MATLAB, LaTeX, Linux, Git/GitHub

Data & Analysis: Stellar Spectroscopy, Differential Abundance Analysis, Atmospheric Parameter Generation, FITS Data Processing

Hardware & Pipelines: Telescope Operation, BACCHUS, Astronomical Image Processing